A Lean Transcription of the Corrected Lazard Formula for Solvable Quintics

ProveIt project

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Abstract

Daniel Lazard gave an explicit radical formula for every irreducible solvable quintic. This note documents a Lean 4 transcription of the formula, including Vladimir Reshetnikov's correction of two terms in the printed expression P_{22} and Valeriy Zapodovnikov's restoration of the exceptional sign choice needed when the first determination of ε would make $T = 0$.

The Lean development contains two interfaces. The older, reusable proof-carrying interface accepts four Fourier relations, derives the five Vieta relations from them, and proves the root and factorization endgames. The newer root-origin development constructs the ordered roots, derives the Fourier and Vieta identities from those roots, proves the corrected displayed numerator identities, and feeds them into radical reconstruction. A denominator-safe alternate path recovers all four fifth powers through coherent projection branches and then adjoins four fifth roots; this avoids all four printed quotients by the possibly zero invariant E. These newer files still await a fresh complete public dependency build, so their exact status is recorded in the claim crosswalk rather than overstated here.

The Gaussian-rational dispatcher source still exposes the older certificate API and proves, classically, that it returns a finite radical vector exactly when a nonzero input is semantically solvable by radicals. Every returned vector is exhaustive, but this dispatcher specification is noncomputable and is not yet a freshly rebuilt public endpoint. This note also records that two printed terms in P_{22} are erroneous and that the paper's Theorems 3 and 4 are false under its own Definition 1; the corrected formal development proves counterexamples rather than asserting those literal claims.

This is a paused research checkpoint, not a claim that every assertion in Lazard's paper has been formalized. Several printed assertions require correction or are false as stated, and several sound replacement theorems have source proofs whose complete Lean dependency closures have not yet been rebuilt. The dated status section below separates those cases precisely.

Contents

1 Purpose, sources, and claim boundary

The core transcription is

`Algebra/PolynomialFormulas/Lean/PolynomialFormulas/LazardQuintic.lean`.

The generic Fourier soundness proof is in

`Algebra/PolynomialFormulas/Lean/PolynomialFormulas/LazardQuinticFourier.lean`.

The generic Vieta endgame is in

`Algebra/PolynomialFormulas/Lean/PolynomialFormulas/LazardQuinticVieta.lean`,

and the Gaussian-rational dispatcher and completeness specification are in

`Algebra/PolynomialFormulas/Lean/PolynomialFormulas/GaussianQuinticSolver.lean` and `Algebra/PolynomialFormulas/Lean/PolynomialFormulas/GaussianQuinticCompleteness.lean`.

These files describe the intended interface. Some are already on the public import surface and some remain worktree checkpoints awaiting the fresh public build described below; this note states that distinction explicitly rather than inferring it from the presence of a source file. The source materials were reviewed at Smithereens commit `da7026ec1369ade49bb6d964019a4c124fdd5091`. The principal references are:

(1) Lazard's chapter *Solving Quintics by Radicals*, especially the explicit formula on printed pp. 219–222 [1];

(2) Lazard's original Maple worksheet, version 3 (March 2002), distributed with those materials;

(3) Reshetnikov's Wolfram-language transcription and warning about the two printed P_{22} errors [5];

(4) Zapodovnikov's corrected 2025 version on the same page, especially `If[(g+h/F)==0,F=-F]`; and

(5) Dummit's independent solvability resolvent [2], together with Chapman's repair of the specialized-distinctness step [3].

The expanded review also covered every text-readable document under `src/Polynomial` whose path or contents mention "quintic", including the Reshetnikov notebooks, the Wolfram notebooks, the independent `solving-quintics` package, Dummit's paper, and the corpus reports. These additional files provide useful examples and independent checks, but they do not supersede Lazard's formula. In particular, the `solving-quintics` package switches to numerical root matching after the resolvent and is not an exact implementation of the formula documented here.

The present status claim is deliberately limited:

> The reusable kernel-checked interface proves the final root identity from four explicit cyclic Fourier relations and the exact five-factor identity from five explicit Vieta relations. Separate root-origin proof sources derive those relations from the actual ordered roots and reconstruct the whole root tuple; their focused checks and their complete public dependency checks are recorded separately. The worktree also contains a classical specification saying that, for every nonzero Gaussian-rational coefficient tuple, a finite returned vector is equivalent to radical solvability and contains exactly the roots. That dispatcher specification is not yet a freshly rebuilt public theorem and is not an executable decision procedure.

This distinction matters. Lean accepts division by zero as a total field operation, and so a raw expression alone would not faithfully express Lazard's partial algebraic construction. The legacy formalization therefore records every required denominator condition and every shared radical determination in `RadicalCertificate`. The root-origin theorems instead construct the relevant relations from a labelled complete root tuple. The alternate reconstruction adds separate fifth roots for all four nonzero Fourier modes and therefore does not pretend that replacing only the first printed quotient resolves the $E = 0$ case.

2 Mathematical scope

Lazard begins with an irreducible quintic over a field of characteristic different from 2 and 5. His compact expanded formula uses a particular fourth projection whose invertibility is justified when -1 is not a square; this is harmless over $\mathbb{Q}$. The paper also sketches an alternate projection for more general fields, but does not print its expanded formula.

The Lean expressions are polymorphic over a field K with `CharZeroK`. This is a slightly stronger characteristic assumption and ensures that all displayed rational denominators are meaningful. The actual rational front end is made explicit by `RationalInvariantCertificate`: the tuple $(i_4, \ldots, i_8)$ is first certified in $\mathbb{Q}$ and only then mapped to a radical extension. The abstract generic API remains useful for algebraic lemmas.

The same characteristic correction is already necessary in Sect. 4.2. Lazard's global exclusions permit characteristic three, but the printed Cardano display divides by 3 and uses a primitive third root of unity. The paired Lean and Coq counterexamples use the separable irreducible cubic $X^3 - X - 1$ over $\mathbb{F}_3$. More strongly, their generic obstruction theorems show that division by three collapses and that $X^3 - 1 = (X - 1)^3$ in *every* characteristic-three field, so passing to an extension cannot repair the missing primitive root. In the corrected scope, the formalizations transcribe the printed values

$$\frac{s_1}{3} - \frac{p}{s_1}, \qquad \frac{j^2 s_1}{3} - \frac{jp}{s_1}, \qquad \frac{j s_1}{3} - \frac{j^2 p}{s_1}, \quad s_1 = 3\sqrt[3]{-q/2 + \sqrt{p^3/27 + q^2/4}},$$

including their printed order. The cyclotomic equation $j^2 + j + 1 = 0$ is proved equivalent to genuine third-root primitivity when $3 \neq 0$. For an irreducible cubic the new aggregate endpoints no longer ask the caller to certify $s_1 \neq 0$: when $p \neq 0$ it follows from the product of the two Cardano radicands, and when $p = 0$ the proved sign-selection branch uses irreducibility to obtain $q \neq 0$ and chooses the safe square-root sign. The endpoints then derive the compatible second radical, the exact three-linear factorization, pointwise soundness, and all-root exhaustiveness. Existence of the square and cube radicals remains an explicit ambient-field requirement, rather than an unproved total radical operation. The new Lean exact-scope bridge, like the corresponding Coq development, uses only the explicit hypotheses $2 \neq 0$ and $3 \neq 0$. The Lean bridge remains source-level. The focused Coq kernels have accepted `CubicField`, the literal Fourier formula, and the generic and concrete characteristic-three obstruction with closed assumption reports; the enlarged public audit is still pending.

There is an independent characteristic gap in the Sect. 3 alternating example. The paper infers a nonzero discriminant from irreducibility and therefore calls the Vandermonde quadratic resolvent automatically separable. This is false over imperfect fields. Both developments now prove the exact criterion

$$X^2 - \Delta \text{ is separable} \quad \Longleftrightarrow \quad 2 \neq 0 \ \text{ and } \ \Delta \neq 0.$$

They also prove that separability of the source polynomial gives a nonzero derivative resultant, which is the premise irreducibility alone fails to supply; the Lean development additionally bridges a monic positive-degree source to its standard discriminant and the actual discriminant resolvent. Each prover exposes one premise-free theorem packaging the whole counterexample. It is constructed internally over $\mathbb{F}_3(t)$: degree at infinity proves that t is not a cube, hence $X^3 - t$ is irreducible; its derivative and discriminant vanish, so it is inseparable and its alternating resolvent is X^2. Characteristic three still satisfies the paper's printed exclusions of characteristics two and five. Thus the corrected statement requires a separable source polynomial (automatic over a perfect, in particular characteristic-zero, field) and characteristic different from two.

The preceding root-resolvent paragraph in Sect. 4.2 is now transcribed literally as well. Its sole root-side input is an explicit complete labelled root presentation, represented by the three depressed Vieta relations for x_0, x_1, x_2; it does not accept a discriminant, resolvent, Fourier, or action certificate. Both provers define

$$V = (x_0 - x_1)(x_1 - x_2)(x_2 - x_0), \qquad \Delta = 4p^3 + 27q^2$$

and derive $V^2 = -\Delta$ by polynomial algebra. Genuine primitivity of j gives $j^2 + j + 1 = 0$, from which they derive $(j - j^2)^2 = -3$ and hence $((j - j^2)V)^2 = 3\Delta$. These are not supplied certificates: the developments prove the actual two-factor polynomial and evaluation identities for $X^2 + \Delta$ and $X^2 - 3\Delta$. They also prove that simultaneously exchanging j with j^2 and x_1 with x_2 fixes the modified invariant. The paper-facing aggregate retains the explicit hypothesis $3 \neq 0$, matching the corrected Cardano scope. These new modules are publicly registered. The focused Coq module is kernel-green with all six assumption reports closed under the ambient field; the Lean module and both enlarged public audits remain pending.

The nearby printed sentence that S_3 has only one subgroup $A_3 = C_3$ is false on its literal reading: a transposition generates a subgroup of order two. Lean and Coq now contain that explicit counterexample, prove $A_3 = C_3$, and prove the strongest intended correction: this common subgroup is the unique *proper transitive* subgroup of S_3 (indeed by equality, not merely up to conjugacy). The Coq proof is structural—it uses transitivity, Lagrange, Cauchy, parity, and cardinality rather than enumerating all subgroups—and is now kernel-green with closed assumption reports. The Lean module and both public audits remain pending.

There is a separate modular issue in the paper's arbitrary-degree Theorem 2. The paper carries its global assumption "characteristic different from 2 and 5" into a statement for every

d and every subgroup of S_d, while the proof averages by the group order. That generality is false: permutation invariant rings need not be Cohen–Macaulay, hence need not be free over the symmetric parameters, in modular characteristic [4]. A small finite counterexample is the regular C_6 action on six variables over $\mathbb{F}_3$. In degree seven, Lean constructs 17 linearly independent classes in the quotient by the positive-degree elementary symmetric parameters, while the Molien/Hilbert calculation forces coefficient 16 under a homogeneous freeness assumption. Thus the contradiction needs only quotient dimension at least 17; the stronger diagnostic assertion that the dimension is exactly 17 is not a formal theorem. The corrected generic endpoints deliberately assume the sufficient nonmodular hypothesis that the subgroup order is nonzero in the coefficient field, equivalently that it is invertible. This is the exact hypothesis used by Reynolds averaging, but it is not necessary for every individual modular permutation action. The Lean source proves that the displayed permutation has order six, identifies its orbit sums with the actual fixed homogeneous spaces, derives the forced count 16 from an arbitrary hypothetical bounded homogeneous basis, and uses an explicit seventeen-class dual certificate in the literal quotient. The Coq source proves the same group and semantic identifications, derives the same Hilbert convolution from an arbitrary actual finite-free decomposition, and checks the closed 132-column/rank-115 product matrix with kernel reduction. Neither final contradiction assumes an orbit count, quotient dimension, or Hilbert coefficient. These source chains remain a completion gate only in the sense that their focused and public kernel builds have not yet accepted them; this modular defect does not affect the rational quintic formula or the nonmodular M_5 computation.

Theorem 1 also needs an orientation and labelling correction. The Lean development uses left cosets aG, left multiplication, and the formal rename by a, whose stabilizer is aGa^{-1}. The Coq development uses right cosets Gx, right translation, and the inverse rename $x^{-1} \cdot P$, whose stabilizer is $x^{-1}Gx$. Both prove the same corrected criterion: the fixed displayed containment gives a base-field resolvent root without a separability premise, whereas an arbitrary base-field root of a separable specialized resolvent gives containment only in a conjugate of G. Their paper-facing wrappers derive the root action, equivariance, and specialized orbit factorization from an exact stabilizer and a duplicate-free exact root factorization. The generic specialization is parametrized by an ambient finite Galois extension; the rational-quintic application obtains that structure from the canonical characteristic-zero splitting field rather than accepting a fixed-field or action certificate. The separate rational-function bridges derive fixed-field generation and the ordinary minimal-polynomial identity; those are not caller-provided certificates. On the Lean side the arbitrary-base fixed-field module now also names the derived degree $d!$, the Galois property, and an explicit equivalence from S_d to the complete automorphism group. The Coq side obtains the same conclusions by localizing the proved reverse-Artin basis, constructing a splitting polynomial from its permutation orbits, and identifying the faithful permutation image with the full Galois group. Thus neither symmetric rational-function endpoint accepts finite-extension, Galois-group, fixed-field, generation, action, or minimal-polynomial certificates. Both provers also expose the literal converse: exact relative stabilizer is equivalent to generation of the subgroup fixed field, and a second theorem bundles this with the ordinary orbit-product/minimal-polynomial identity. The reverse proof uses Galois correspondence (respectively `gal_fixedField`) and only the generator equality; the minpoly conjunct is therefore explicitly documented as redundant in that direction. The Coq bridge now transports this converse through the concrete symmetric rational-function presentation, so its endpoint concerns the exact stabilizer of the original formal multivariate polynomial rather than only an abstract Galois element. Lean's source-level direct aggregate `sectionThreeAndTheoremOne_exactInvariantRootCriterion` composes the fixed-field identity, degree, Galois property, ordinary minimal polynomial, fixed-field generation, derived root action, and corrected conjugate-containment criterion in one theorem. The generic criterion, its explicit invariant construction, and the theta adapter are committed and freshly kernel-checked. The universal rational-function minimal-polynomial bridge is a separate module; it does not assert that every specialized theta sextic remains minimal after root substitution. The larger aggregate and the polynomial-realized counterexample remain source-level checkpoints pending their own dependency builds. Beyond the finite-group coset witness, both provers now realize the obstruction on the explicit irreducible depressed cyclic rational quintic. They give a duplicate-free exact ordering and a literal five-factor presentation of its roots in the splitting field, derive the genuine Galois permutation action, and preserve a base-field root and separability of the scalar resolvent under an explicit three-cycle relabelling. The resulting image lies in the selected conjugate of F_{20} but not in the displayed standard F_{20}. Thus the displayed subgroup cannot replace its conjugacy class even for an actual polynomial. The formerly missing named adapter is now present at source level in both provers. Lean's `LazardGeneralResolventThetaAdapter.lean` defines the rational ten-term theta polynomial, proves its exact renaming stabilizer, and proves `paperRootTupleResolvent_map_eq_scalarResolvent`; the realized file then invokes the generic corrected criterion literally in `RealizedFixedSubgroupCounterexample.literal_correctedTheoremOn` with a closed cyclic-quintic bundle. Coq's `LazardGeneralResolventThetaAdapter.v` packages the exponent table as a multivariate polynomial, proves its exact stabilizer, makes the forced inverse-representative right-coset orientation explicit, and proves `lazard_paper_ordered_theta_resolvent_map_eq_scalar_resolvent`. The realized Coq file constructs the required base and splitting subfields, transports the scalar root and separability, and invokes the generic corrected theorem in `cyclic_paper_theta_resolvent_has_root_iff_image_sub_conj` The Lean theta adapter and finite coset counterexample are kernel-green. The polynomial realization has the separate source-checkpoint status recorded below because it still depends on the unfinished Dummit certificate closure.

The older propositions `FormulaSound` and `FormulaFactors` quantify over an arbitrary characteristic-zero field after the displayed invariant and radical certificates have been supplied. Audit showed that these hypotheses still admit extraneous invariant tuples, so the propositions are over-strong: they are retained definitions, not theorems or valid proof targets. The sound legacy certificate interface consists of `FourierRelations`, whose four cyclic equations are the additional evidence used by `solveGeneral_root_of_fourierRelations`. The generic development derives `FiveRootRelations` from those equations; its five elementary-symmetric conclusions imply an exact factorization. The Gaussian `LazardWitness` stores the Fourier family, not an independent Vieta certificate. Lazard's sourced irreducibility and -1 nonsquare hypotheses remain relevant to deriving that evidence and to the applicability/existence bridge. In the newer root-origin path, the relation families are intermediate theorem conclusions rather than caller-provided correctness assumptions.

An executable complete solver on arbitrary quintics would need to factor reducible inputs and search for the rational root of the sextic. Every proper factor has degree at most four and can in principle be handled by the existing low-degree formulas. The Lazard core implements neither search: its rational entry point accepts a supplied `RationalInvariantCertificate`. The Gaussian dispatcher covers arbitrary solvable inputs only as a classical specification, using a chosen `CompleteRadicalSolution` when no Lazard witness is available. It therefore does not misrepresent Lazard's irreducible branch as an executable total reducer.

3 The algorithm at a glance

For an irreducible rational quintic, the corrected construction is:

1. normalize and translate to $Y^5 + pY^3 + qY^2 + rY + s$;

2. find the rational root i_4 of Lazard's sextic resolvent;

3. solve the four linear equations of Figure 3 for i_5, i_6, i_7, i_8;
4. calculate D, E, F, G and choose $\varepsilon^2 = 5D$;
5. if $E + F/\varepsilon = 0$, replace ε by $-\varepsilon$;
6. choose $T^2 = \frac{5}{2}(E + F/\varepsilon)$ and set $U = 5G/(T\varepsilon)$;
7. among four coherent sign triples (ε, T, U), choose one for which $Q_1 \neq 0$;
8. choose one shared fifth root $P_1^5 = Q_1$, calculate the corrected P_2, P_3, P_4, and apply the inverse discrete Fourier transform;
9. translate every resulting root back by $-b/(5a)$.

Step 8 is the compact printed branch and requires the standard projection denominator E to be nonzero. When it is zero, the formal fallback keeps the same quadratic data but uses the corrected coherent projection matrix to recover $s_1^5, s_2^5, s_4^5, s_3^5$ separately. It adjoins a fifth root of each value and reconstructs all roots directly by inverse Fourier transform. No claim is made that this larger explicit tower is minimal.

The group-theoretic tower behind these steps is

$$C_5 \subset D_5 \subset F_{20} \subset S_5, \qquad [D_5 : C_5] = 2, \quad [F_{20} : D_5] = 2, \quad [S_5 : F_{20}] = 6.$$

The sextic selects a conjugate of the maximal solvable subgroup F_{20}; the two index-two inclusions explain two square layers, and Kummer theory explains the terminal cyclic degree-five layer once a primitive fifth root is present. This field-theoretic presentation is distinct from the denominator-safe formula implementation, which may adjoin four explicitly chosen fifth roots without asserting minimality.

4 Normalization and depression

Write

$$f_0(X) = aX^5 + bX^4 + cX^3 + dX^2 + eX + f, \qquad a \neq 0.$$

Under $X = Y - b/(5a)$ and division by a, one obtains

$$f(Y) = Y^5 + pY^3 + qY^2 + rY + s,$$

where

$$\begin{aligned}
p &= \frac{5ac - 2b^2}{5a^2}, \\
q &= \frac{25a^2d - 15abc + 4b^3}{25a^3}, \\
r &= \frac{125a^3e - 50a^2bd + 15ab^2c - 3b^4}{125a^4}, \\
s &= \frac{3125a^4f - 625a^3be + 125a^2b^2d - 25ab^3c + 4b^5}{3125a^5}.
\end{aligned}$$

These are `depress`. Lean proves, for every Y,

$$f_0\left(Y - \frac{b}{5a}\right) = a\, f(Y)$$

as `depress_eval`. It also proves that depression commutes with a field homomorphism (`depress_map`), which is the bridge from rational certificates to the radical field.

5 Fourier invariants and metacyclic data

Let $x_0, \dots, x_4$ be an ordering of the roots and let ω be a primitive fifth root of unity. Lazard uses the Fourier sums

$$s_k = \sum_{j=0}^{4} \omega^{kj} x_j, \qquad k = 0, \dots, 4.$$

The depressed condition gives $s_0 = 0$. Define cyclic invariants

$$S_1 = s_1^5, \qquad S_2 = s_2 s_1^3, \qquad S_3 = s_3 s_1^2, \qquad S_4 = s_4 s_1.$$

Once $P_1^5 = S_1$ is chosen with $P_1 \neq 0$, the remaining Fourier components are $P_2 = S_2/P_1^3$, $P_3 = S_3/P_1^2$, and $P_4 = S_4/P_1$.

There is a sign error in the printed combined-action discussion. If $\phi(x_i) = x_{2i}$, the displayed definitions give

$$\phi(T) = -U_{\text{printed}}, \qquad \phi(U_{\text{printed}}) = T,$$

not $\phi(U_{\text{printed}}) = -T$. Figure 3 instead uses the convention $U_{\text{formula}} = -U_{\text{printed}}$, for which $\phi(U_{\text{formula}}) = -T$. The Lean and Coq developments keep those two conventions distinct. They prove that the literal forward source

$$[P_1^5, P_3^5, P_4^5, P_2^5]$$

is the actual four-term ϕ-orbit, and that the formula-convention projection tuple is exactly $[I_1, I_2, I_3, -I_4]$. They also prove the commutation of ϕ with $\psi(\omega) = \omega^2$, the resulting fixedness, and independence from the choice of primitive fifth-root generator. Both focused modules have been accepted by their kernels; rebuilding the enlarged public aggregate audits remains a separate completion gate.

The rational metacyclic basis used to express these quantities is

$$i_4 = \sum_C x_0^2(x_1x_4 + x_2x_3), \qquad i_5 = \sum_C x_0^3(x_1x_4 + x_2x_3),$$
$$i_6 = \sum_C x_0^4(x_1x_4 + x_2x_3), \qquad i_7 = \sum_C x_0^3(x_1^2x_4^2 + x_2^2x_3^2),$$
$$i_8 = \sum_C x_0^4(x_1^2x_4^2 + x_2^2x_3^2).$$

Lean stores their values in `Invariants`. The exact resolvent and linear relations do not by themselves certify that an arbitrary supplied tuple came from roots. The root-origin modules close that distinction by constructing the tuple from an ordered complete root presentation and deriving the Fourier relations from the resulting Fourier sums. The public audit still checks the entire dependency chain separately from the generic certificate endgame.

6 The sextic resolvent

The discriminant of $X^5 + pX^3 + qX^2 + rX + s$ is

$$\begin{aligned}\Delta = {} & 108p^5s^2 - 72p^4qrs + 16p^4r^3 + 16p^3q^3s - 4p^3q^2r^2 - 900p^3rs^2 \\ & + 825p^2q^2s^2 + 560p^2qr^2s - 128p^2r^4 - 630pq^3rs + 144pq^2r^3 \\ & - 3750pqs^3 + 2000pr^2s^2 + 108q^5s - 27q^4r^2 + 2250q^2rs^2 \\ & - 1600qr^3s + 256r^5 + 3125s^4.\end{aligned}$$

This is `discriminant`. The theorem `discriminant_eq_polynomial_discr` proves symbolically, for every depressed quintic in the stated characteristic-zero field scope, that this display is Mathlib's polynomial discriminant; the proof is not a finite sample comparison.

For a variable $z = i_4$, put

$$\begin{aligned} A(z) &= 2z^3 + 8rz^2 + (-6p^2r + 2pq^2 - 50qs + 24r^2)z \\ &\quad - 15p^2qs - 16p^2r^2 + 13pq^2r + 125ps^2 - 2q^4 - 200qrs + 64r^3. \end{aligned}$$

Then Lazard's monic sextic is

$$R(z) = \left(\frac{A(z)}{2}\right)^2 - \left(z + 3r + \frac{p^2}{4}\right)\Delta. \tag{1}$$

The fraction-free equation used by the corrected Wolfram code is

$$A(z)^2 = (4z + p^2 + 12r)\Delta. \tag{2}$$

Lean's two definitions are `resolventEval` and `resolventPolynomial`. Their agreement under evaluation is the theorem `resolventPolynomial_eval`. The theorem `resolventEval_eq_zero_iff` proves the equivalence of (1) and (2), while `resolventPolynomial_monic` proves literal monicity before any comparison with a scaled executable resolvent.

There is a notation trap in the printed sources. Cayley's earlier invariant is

$$\Theta = (V - W)^2 = 4i_4 + p^2 + 12r,$$

whereas the section-7 sextic variable is $z = i_4$. Lean uses separate names `cayleyTheta` and `resolventCore`, preventing this collision.

The source theorem `resolventPolynomial_has_rational_root_iff_gal_isSolvable` states that, for an irreducible depressed rational quintic, the sextic has a rational root if and only if its Galois group is solvable. The generic theta orbit product is freshly proved equal, after scalar extension, to Dummit's independent F_{20} sextic. The specialized coefficient theorem still depends on the unfinished fresh rebuild of the Dummit certificate closure, so this note does not yet promote the rational-root equivalence to the current public kernel-checked boundary. Specialized distinctness must not be inferred from generic transitivity alone. The new field-generic Chapman endpoints derive it from $5 \neq 0$, irreducibility, the base-field root sum, and an explicit five-cycle endomorphism, and conclude separability of the six-factor scalar resolvent. Their new splitting-field specializations now derive those hypotheses for an arbitrary irreducible degree-five polynomial. Degree five and $5 \neq 0$ make the derivative nonzero, hence the polynomial separable; irreducibility makes the Galois action on the five roots transitive; and Cauchy's theorem supplies an order-five element. After an existential reindexing, that element acts as the literal standard five-cycle used by Chapman's argument. Lean uses the canonical `p.SplittingField`; Coq states the corresponding `splittingFieldFor` generation hypothesis explicitly. These root orderings and automorphisms are noncanonical witnesses, not executable selection functions. Both expanded specializations remain source-level until their focused and public kernel builds accept them.

7 Recovering $i_5, \ldots, i_8$

After choosing i_4, Lazard's Figure 3 gives four equations linear in $i_5, \ldots, i_8$. They are transcribed as `i4SquareRhs`, `i4CubeRhs`, `i4FourthRhs`, and `i4FifthRhs`; together with the sextic equation they form `InvariantRelations`.

$$i_4^2 = 5i_8 - 2pi_6 + 4qi_5 - 2p^2i_4 - 6p^2r + 2pq^2 + 10qs + 4r^2,$$

$$
\begin{aligned}
i_4^3 = \frac{1}{2}\big[&(3p^2-20r)i_8+(-pq-50s)i_7+(-3p^3+28pr-12q^2)i_6\\
&+(3p^2q-45ps-6qr)i_5+(-3p^4+36p^2r-15pq^2+60qs-32r^2)i_4\\
&-6p^4r+3p^3q^2+41p^2qs+52p^2r^2-54pq^2r-250ps^2+14q^4+140qrs-80r^3\big],\\
i_4^4 = {}&(19p^2r-9pq^2+225qs-60r^2)i_8+(15p^2s-8pqr+3q^3+100rs)i_7\\
&+(-4p^3r+4p^2q^2-105pqs-16pr^2+29q^2r+125s^2)i_6\\
&+(-9p^3s+17p^2qr-8pq^3+140prs+155q^2s-68qr^2)i_5\\
&+(-4p^4r+4p^3q^2-79p^2qs-16p^2r^2+15pq^2r-25ps^2+4q^4+80qrs)i_4\\
&+6p^4qs-22p^4r^2+16p^3q^2r-4p^2q^4-404p^2qrs+68p^2r^3\\
&+132pq^3s+42pq^2r^2+550prs^2-30q^4r-50q^2s^2+20qr^2s+16r^4,
\end{aligned}
$$

and

$$
\begin{aligned}
2i_4^5 = {}&(15p^4r-5p^3q^2+290p^2qs-152p^2r^2-27pq^2r-1375ps^2+22q^4-700qrs+240r^3)i_8\\
&+(18p^4s-11p^3qr+3p^2q^3-530p^2rs+110pq^2s+124pqr^2-41q^3r-2375qs^2+200r^2s)i_7\\
&+(-15p^5r+5p^4q^2-212p^3qs+168p^3r^2-83p^2q^2r+325p^2s^2+10pq^4+1560pqrs\\
&\qquad-176pr^3-620q^3s-12q^2r^2-1500rs^2)i_6\\
&+(15p^4qr-5p^3q^3-147p^3rs+351p^2q^2s-90p^2qr^2-43pq^3r-3175pqs^2\\
&\qquad-420pr^2s+20q^5+215q^2rs+152qr^3+625s^3)i_5\\
&+(-15p^6r+5p^5q^2-200p^4qs+200p^4r^2-110p^3q^2r+355p^3s^2+15p^2q^4\\
&\qquad+1728p^2qrs-432p^2r^3-752pq^3s+220pq^2r^2-200prs^2-43q^4r+1825q^2s^2\\
&\qquad-2640qr^2s+512r^4)i_4\\
&-30p^6r^2+25p^5q^2r+198p^5s^2-5p^4q^4-491p^4qrs+364p^4r^3+181p^3q^3s\\
&-286p^3q^2r^2-810p^3rs^2+95p^2q^4r+3005p^2q^2s^2+4120p^2qr^2s-1088p^2r^4\\
&-12pq^6-4095pq^3rs+612pq^2r^3-15875pqs^3+900pr^2s^2+858q^5s-34q^4r^2\\
&+10700q^2rs^2-6240qr^3s+960r^5\\
&\qquad+6250s^4.
\end{aligned}
$$

Retaining these as equations, instead of embedding the enormous pre-solved rational functions, is intentional. It is exactly the finite linear-system interface needed for a later proof of existence and uniqueness.

8 The two quadratic stages

The explicit metacyclic invariants on Lazard p. 221 are:

$$
\begin{aligned}
D = {}&40pi_8-120qi_7+(-24p^2+100r)i_6+(88pq-300s)i_5\\
&+(-24p^3+100pr+24q^2)i_4-80p^3r+40p^2q^2-480pqs+160pr^2+332q^2r+125s^2,\\
E = {}&(3p^2+20r)i_6+(-pq-50s)i_5+(3p^3+12pr+3q^2)i_4\\
&+4p^3r-3p^2q^2+40pqs+16pr^2-21q^2r+125s^2,\\
F = {}&(-65p^2q+875ps-550qr)i_8+(-58p^2r+41pq^2-275qs+440r^2)i_7\\
&+(85p^3q-520p^2s-298pqr+366q^3+2100rs)i_6\\
&+(4p^3r-73p^2q^2+2095pqs-56pr^2-748q^2r-4875s^2)i_5\\
&+(85p^4q-418p^3s-440p^2qr+419pq^3+1590prs-1040q^2s+524qr^2)i_4\\
&-12p^5s+158p^4qr-85p^3q^3-1462p^3rs-159p^2q^2s+142p^2qr^2
\end{aligned}
$$

$$
\begin{aligned}
&+ 896pq^3r + 175pqs^2 + 2900pr^2s - 402q^5 - 1925q^2rs - 448qr^3 - 1875s^3,\\
G &= (-35p^2q - 250ps - 200qr)i_8 + (-22p^2r + 19pq^2 + 650qs - 40r^2)i_7\\
&+ (15p^3q + 195p^2s + 68pqr - 6q^3 - 1100rs)i_6\\
&+ (-4p^3r - 27p^2q^2 - 270pqs + 96pr^2 - 182q^2r + 3000s^2)i_5\\
&+ (15p^4q + 213p^3s + 50p^2qr + pq^3 - 940prs + 515q^2s - 184qr^2)i_4\\
&+ 12p^5s + 42p^4qr - 15p^3q^3 + 492p^3rs - 156p^2q^2s + 358p^2qr^2\\
&- 246pq^3r + 2825pqs^2 - 1400pr^2s + 42q^5 + 550q^2rs - 232qr^3 - 1250s^3.
\end{aligned}
$$

They are `invariantD`, `invariantE`, `invariantF`, and `invariantG`.

Choose ε_0 with $\varepsilon_0^2 = 5D$. The corrected choice is

$$
\varepsilon = \begin{cases} -\varepsilon_0, & E + F/\varepsilon_0 = 0,\\ \varepsilon_0, & E + F/\varepsilon_0 \neq 0. \end{cases} \tag{3}
$$

This is literally `correctEpsilon`. Lean proves `correctEpsilon_sq`. Under the sourced assertion that at least one of $E + F/\varepsilon_0$ and $E - F/\varepsilon_0$ is nonzero, the theorem

`correctedTRadicand_ne_zero`

proves that the corrected radicand is nonzero. One then chooses

$$
T^2 = \frac{5}{2}\left(E + \frac{F}{\varepsilon}\right), \qquad U = \frac{5G}{T\varepsilon}. \tag{4}
$$

The factor $5/2$ is the one used in section 7, Maple, and both Wolfram ports; an earlier prose passage in Lazard omits the relevant scaling.

The records `EpsilonChoice`, `TChoice`, and `RadicalCertificate` expose the equations and nondegeneracy assumptions. In particular, the certificate stores the complementary equation $U^2 = \frac{5}{2}(E - F/\varepsilon)$. The theorem `RadicalCertificate.t_nonzero` derives $T \neq 0$ from the corrected radicand rather than trusting total field division, and `RadicalCertificate.initial_relations` proves $TU\varepsilon = 5G$ from the definition of U.

9 The fifth-root radicand and coherent signs

Define

$$
\begin{aligned}
H &= 25(2i_5 - pq - 5s),\\
I &= 25\big[40pi_8 - 70qi_7 + (-24p^2 + 100r)i_6 + (68pq - 300s)i_5\\
&\qquad + (-24p^3 + 100pr - 46q^2)i_4 - 80p^3r + 20p^2q^2 - 255pqs + 160pr^2 - 28q^2r + 125s^2\big],\\
J &= -25pi_8 - 25qi_7 + (-9p^2 - 60r)i_6 + (-7pq + 525s)i_5\\
&\quad + (-p^3 - 96pr + 11q^2)i_4 + 50p^3r - 7p^2q^2 - 145pqs - 308pr^2 + 128q^2r - 1000s^2,\\
K &= -125pi_8 + 75qi_7 + (67p^2 - 420r)i_6 + (-109pq + 1175s)i_5\\
&\quad + (63p^3 - 412pr + 27q^2)i_4 + 210p^3r - 79p^2q^2 - 415pqs - 676pr^2 + 496q^2r - 750s^2.
\end{aligned}
$$

These are `invariantH` through `invariantK`. For a coherent triple (ε, T, U), set

$$
Q_1 = \frac{5}{4}\left(H + \frac{I}{\varepsilon} + \frac{TJ + UK}{E}\right). \tag{5}
$$

The four allowed triples are

$$
(\varepsilon, T, U), \quad (\varepsilon, -T, -U), \quad (-\varepsilon, U, -T), \quad (-\varepsilon, -U, T).
$$

They are the four constructors handled by `branchTriple`. The selector `selectSignBranch` returns the first branch with $Q_1 \neq 0$, or `none` if the mathematical existence hypothesis has not been supplied. Its theorem `selectSignBranch_sound` proves that a returned branch is indeed nonzero. Quadratic coherence is the separate theorem

`QuadraticRelations.of_branchTriple`.

It proves that each transformation preserves both square equations and $TU\varepsilon = 5G$. A `RadicalCertificate` stores the chosen branch, the nonzero fact, and one shared P_1 satisfying $P_1^5 = Q_1$; Lean derives $P_1 \neq 0$ in `RadicalCertificate.p1_nonzero`.

This finite branch mechanism is the algebraic content of the corrected algorithm. Domen's later use of Mathematica's `Together` merely repairs syntactic evaluation of fractional powers during root matching. Since Lean uses a single ω and field expressions, it has no mathematical analogue here.

10 The corrected P_{22} and Fourier reconstruction

Lazard's remaining numerators are transcribed one per Lean definition:

$$\begin{aligned}
P_{41} &= -5p,\\
P_{42} &= 5(10i_7 - 4pi_5 - 14qi_4 - 4p^2q + 45ps - 72qr),\\
P_{31} &= -25q,\\
P_{32} &= 25(-10i_8 + 2pi_6 - 22qi_5 + 2p^2i_4 + 20p^2r + 2pq^2 - 35qs - 40r^2),\\
P_{33} &= 5(35i_8 - 4pi_6 + 23qi_5 + (-6p^2 + 12r)i_4 - 58p^2r + 14pq^2 - 105qs + 76r^2),\\
P_{34} &= 5(5i_8 - 22pi_6 + 14qi_5 + (-18p^2 + 16r)i_4 - 34p^2r + 22pq^2 - 140qs + 68r^2),\\
P_{21} &= 5(3i_4 + 2p^2 - 16r),\\
P_{22} &= 25\big(-10qi_6 + (8p^2 - 50r)i_5 + (-2pq - 25s)i_4\\
&\qquad\qquad + \boxed{8p^3q} + \boxed{70q^3} - 20p^2s - 26pqr + 50rs\big),\\
P_{23} &= 25(-4pi_7 - qi_6 + 4ri_5 + (-3pq + 15s)i_4 + 26p^2s - 26pqr + 7q^3 - 40rs),\\
P_{24} &= 25(3pi_7 - 18qi_6 + 22ri_5 + (-14pq + 20s)i_4 + 18p^2s - 33pqr + 21q^3 + 30rs).
\end{aligned}$$

The boxed terms are Reshetnikov's corrections. Lazard p. 222 prints $8p^3$ and $70q^3q$; the corrected terms are independently confirmed by the original Maple worksheet, which has $200p^3q$ and $1750q^3$ after distributing the outer factor 25. Lean makes the corrected formula conspicuous in `p22`.

With the selected (ε, T, U) and shared P_1, define

$$\begin{aligned}
P_4 &= \frac{P_{41}}{2P_1} + \frac{P_{42}}{2\varepsilon P_1},\\
P_3 &= \frac{P_{31}}{4P_1^2} + \frac{P_{32}}{4\varepsilon P_1^2} + \frac{P_{33}T + P_{34}U}{10EP_1^2},\\
P_2 &= \frac{P_{21}}{4P_1^3} + \frac{P_{22}}{4\varepsilon P_1^3} + \frac{P_{23}T + P_{24}U}{10EP_1^3}.
\end{aligned}$$

These are `fourierP4`, `fourierP3`, and `fourierP2`.

For one primitive fifth root ω, all five depressed candidates are

$$y_k = \frac{\omega^k P_1 + \omega^{2k} P_2 + \omega^{3k} P_3 + \omega^{4k} P_4}{5}, \qquad k = 0, 1, 2, 3, 4. \tag{6}$$

The translated candidate values are

$$x_k = y_k - \frac{b}{5a}.$$

Lean represents ω by `FifthRootOfUnity`; its field `primitive` certifies that `value` is a primitive fifth root. The single indexed formula `solveDepressed` replaces five branch-sensitive Mathematica expressions. `solveGeneral` performs the final translation, and `solveRational` keeps the rational invariant certificate visible. In the reusable interface, when the four `FourierRelations` hold, the proved theorem `solveGeneral_root_of_fourierRelations` establishes that every x_k annihilates the original quintic, and `solveGeneral_fiveRootRelations` derives `FiveRootRelations` for these five concrete values. Their five conclusions are the scaled Vieta identities

$$b = -a\sum_k x_k, \qquad c = a\sum_{j<k} x_j x_k, \qquad d = -a\sum_{i<j<k} x_i x_j x_k,$$
$$e = a\sum_{i<j<k<\ell} x_i x_j x_k x_\ell, \qquad f = -a\prod_k x_k.$$

The theorem `FiveRootRelations.factorization` expands those compact identities and proves in the kernel that

$$aX^5 + bX^4 + cX^3 + dX^2 + eX + f = a\prod_{k=0}^{4}(X - x_k).$$

Thus `LazardWitness.factorization` preserves multiplicities and proves exhaustiveness when $a \neq 0$; the five identities are not supplied as witness fields. The separate root-origin chain also derives the required Fourier premise from the actual root tuple: inverse Fourier reconstruction identifies every output with a reordered root, and root-derived elementary symmetric identities supply the Vieta factorization. Those source modules must still pass the final public builds before this note calls the integration complete. The Coq root-origin bridge now exposes the same composition by name: `lazard_root_fourier_vieta`, `lazard_root_fourier_eval_factorization`, `lazard_root_fourier_output_root`, `lazard_root_fourier_output_complete`, and `lazard_root_fourier_output_root_iff`. Their only root-side input is the actual ordered tuple together with its depressed sum equation; no Fourier, Vieta, factorization, or correctness certificate is accepted from the caller.

11 Lean-to-paper crosswalk

Lean declaration	Lazard notation/source	Role
`GeneralQuintic`, `GeneralQuintic.polynomial`	a, b, c, d, e, f	Original polynomial and its coefficient representation.
`GeneralQuintic.polynomial_eval`	evaluation at x	Proved agreement of polynomial and scalar evaluation.
`DepressedQuintic`, `depress`	p, q, r, s; p. 219	Normalize and depress.
`depress_eval`	$X = Y - b/(5a)$	Kernel-checked translation identity.
`discriminant`	Δ; pp. 219–220	Explicit depressed-quintic discriminant.
`resolventCore`	$A(z)$	Cubic whose square enters the sextic.

Lean declaration	Lazard notation/source	Role
`cayleyTheta`	$\Theta = 4i_4 + p^2 + 12r$; p. 219	Keeps Cayley Θ distinct from $z = i_4$.
`resolventPolynomial`, `resolventPolynomial_monic`	$R(z)$; pp. 219–220	Literal monic degree-six polynomial.
`resolventEval_eq_zero_iff`	fraction-free Stack Exchange equation	Proved equivalence of the two displays.
`Invariants`	$i_4, \ldots, i_8$; pp. 217–220	Metacyclic invariant values.
`i4SquareRhs`–`i4FifthRhs`	Figure 3; pp. 220–221	Linear system for $i_5, \ldots, i_8$.
`InvariantRelations`, `InvariantRelations.map`	sextic plus Figure 3	Exact rational-stage certificate, proved stable under scalar extension.
`invariantD`–`invariantG`	D, E, F, G; p. 221	Quadratic-stage invariant polynomials.
`correctEpsilon`	$\varepsilon \mapsto -\varepsilon$ if $T = 0$; p. 221 and Zapodovnikov	Corrected first sign choice.
`squareBranchAvailable`	not both $E \pm F/\varepsilon_0$ zero	Explicit nondegeneracy obligation.
`correctedTRadicand_ne_zero`	p. 221 nonvanishing argument	Proved consequence of the obligation.
`radicalU`	$U = 5G/(T\varepsilon)$	Second quadratic-stage value.
`invariantH`–`invariantK`	H, I, J, K; pp. 221–222	Projections defining Q_1.
`QuadraticRelations`	equations for ε, T, U	Records both squares and $TU\varepsilon = 5G$.
`SignBranch`, `branchTriple`	four coherent triples	Finite shared-sign orbit.
`QuadraticRelations.of_branchTriple`	coherent sign action	Proves all four transformations preserve the equations.
`selectSignBranch`	paper's $Q_1 \neq 0$ choice	Honest `Option`; no unsafe list indexing.
`q1`	Q_1	Fifth-root radicand.
`p22`	corrected P_{22}; p. 222/MSE	Contains $8p^3q$ and $70q^3$.
`fourierP2`–`fourierP4`	P_2, P_3, P_4; p. 222	Remaining Fourier components.
`RadicalCertificate`	shared ε, T, U, P_1	Equations and nonzero obligations stored once.
`RadicalCertificate.initial_relations`, `RadicalCertificate.chosen_relations`	quadratic coherence	Connect the certificate to both the initial and selected triples.
`FifthRootOfUnity`	primitive ω	Shared fifth root of unity.
`solveDepressed`	inverse Fourier formula	All five depressed candidates.
`solveGeneral`	$x_k = y_k - b/(5a)$	Undo translation.
`RationalInvariantCertificate`	rational $i_4, \ldots, i_8$	A supplied witness over $\mathbb{Q}$; it is not a search procedure.
`FourierRelations`	four cyclic coefficient identities	Formula-specific soundness evidence.
`FourierRelations.inverseFourier_root`	inverse Fourier root identity	Proved generic algebraic theorem.
`solveGeneral_root_of_fourierRelations`	translated root identity	Proved pointwise soundness from `FourierRelations`.
`FiveRootRelations`	five scaled Vieta identities	Formula-independent coefficient certificate for five displayed values.
`FiveRootRelations.factorization`	$f = a\prod_{k=0}^{4}(X - x_k)$	Exact generic factorization derived from the Vieta certificate.

Lean declaration	Lazard notation/source	Role
`LazardWitness`	invariant, radical, and Fourier evidence	Conditional proof-carrying invocation; Vieta relations and factorization are derived rather than stored as witness fields.
`LazardWitness.factorization`	exact product for formula outputs	Multiplicity-preserving specialization of the generic Vieta theorem.
`rootFourierNumeratorRelations`, root reconstruction modules	root-defined P_{ij} and Fourier sums	Derive the formula relations from an actual ordered root tuple rather than accepting them from a caller.
`rootCoherentAlternate_radical_contains_and_reconstructs`	denominator-safe alternate projection	Four coherent recoveries, four fifth adjunctions, and exact inverse-Fourier reconstruction; source-level pending public audit.
`rootCoherentAlternateFourierCertificate_eval_eq_zero_iff`	alternate-output correctness	Root-derived Vieta data, exact factorization, pointwise soundness, and exhaustion with no $E \neq 0$ premise; source-level pending public audit.
`exists_general_commonCompositum_coherentAlternate_completeRootVector`	corrected general rational wrapper	Translates the common-compositum alternate outputs to an arbitrary nondegenerate irreducible rational quintic and returns radical membership, injectivity, exact factorization, and root-set equivalence; source-level pending public audit.
`CompleteRadicalSolution`	five radical expressions plus product identity	Classical fallback certificate for genuine quintics.
`Result.rootVector?`	optional finite output vector	Uniform view of delegated, Lazard, and fallback results.
`solve_hasRootVector_iff_completelySolvableByRadicals`	semantic existence equivalence	For nonzero Gaussian-rational inputs; classical and noncomputable.
`solve_eval_eq_zero_iff_returnedVector_contains`	exact root membership	Soundness and exhaustiveness of every returned vector.
`cyclicLazard_theoremThree_counterexample`; Coq `lazard_theorem3_no_isomorphic_subextension_counterexample`	printed Theorem 3	Source-level unconditional cyclic-quintic counterexample: the polynomial is monic and irreducible, the literal radical competitor contains every root (the Coq package also exposes its exact linear-factor product), the actual formula field and competitor contain the same selected root, and the radical formula field has degree 20 but admits no rational algebra embedding into the degree-10 competitor. Both provers also negate the literal existence of an isomorphic radical subextension; all endpoints are registered for the complete public audit build.

Lean declaration	Lazard notation/source	Role
`lazard_theorem4_counterexample`; Coq `lazard_theorem4_counterexample`	printed Theorem 4	Source-level unconditional irreducible cyclic-quintic/all-roots counterexample: a literal Definition 1 radical extension contains every root but not the primitive fifth root, refuting the printed leastness claim. Coq extracts the proved factor list, proves the universal property of its all-root splitting field, adjoins the displayed primitive fifth root, and proves that exact generated field is not least; it also retains the stronger uniform leastness refutation. All endpoints are registered for the complete public audit build.
`exists_solvable_F20_commonCompositum_exact_paper_count`; Coq `exists_lazard_actual_F20_kummer_common_compositum_compressed`	corrected degree/radical-count part of Theorem 4	From only irreducibility, degree five, and solvability of the actual Galois group, the source-level endpoints derive the F_{20} subgroup tower and explicitly return $\lvert\,\mathrm{Gal}\,\rvert = 5 \cdot 2^{e_{\mathbb{Q}}}$ with its normal order-five subgroup. They construct a primitive fifth root in the product splitting field, prove and return its primitivity and containment in the exact two-square field $\mathbb{Q}(\omega)$, distinguish the compressed exponent $e_\omega \le e_{\mathbb{Q}}$, identify the top field with the embedded quintic splitting field joined with $\mathbb{Q}(\omega)$, and prove radicality, degree $5 \cdot 2^{2+e_\omega}$, relative Galois order $5 \cdot 2^{e_\omega}$, and an exact $2 + e_\omega$ square/one-fifth presentation. Coq derives the formerly explicit nonzero-polynomial premise from the quintic-size hypothesis. Kernel checking of these expanded endpoints remains pending.
`FormulaSound`, `RationalFormulaSound`	raw pointwise claim	Over-strong retained propositions; not theorems.
`FormulaFactors`, `RationalFormulaFactors`	raw five-factor claim	Over-strong retained propositions; not theorems.

12 Why the legacy certificates are explicit

The following denominators occur in the formula:

$$a, \qquad \varepsilon, \qquad T, \qquad E, \qquad P_1.$$

The formalization treats their nonvanishing as follows.

- $a \neq 0$ is an explicit hypothesis of `depress_eval` and `solveGeneral_root_of_fourierRelations`.

- The field `epsilon0_nonzero` records the sourced irreducible-case fact; the theorem `epsilon_nonzero` in the `RadicalCertificate` namespace proves that sign correction preserves it.

- `epsilon_branches`, the square equation for T, and `correctedTRadicand_ne_zero` imply $T \neq 0$.

- The stored square equation for U, together with the definition $U = 5G/(T\varepsilon)$, yields `RadicalCertificate.initial_relations`; `QuadraticRelations.of_branchTriple` proves that all four coherent sign choices preserve the two square equations and $TU\varepsilon = 5G$.

- `invariantE_nonzero` records Lazard's nonvanishing argument; for the formula-stage variables the proved normalization is $T^2 + U^2 = 5E$.

- `branch_nonzero` and $P_1^5 = Q_1$ imply $P_1 \neq 0$.

The printed formulas for P_1, P_2, P_3, P_4 all use the standard projection denominator E. Consequently, replacing only the P_1 row cannot justify the case $E = 0$. The corrected fallback instead uses one coherent invertible projection vector to recover each of the four Fourier fifth powers under a matching sign branch, then adjoins four fifth roots and applies inverse Fourier reconstruction. It is a larger radical tower, but it has no hidden division by E. The explicit Gaussian-integer tuple in `LazardQuinticActualRootEZeroExample` has five distinct depressed roots, nonzero root epsilon, and coefficient invariant $E = 0$; on that tuple the standard denominator and matrix determinant are zero while the coherent-alternate denominator and matrix determinant are nonzero. The example now also specializes the root-derived Vieta chain to an exact alternate-output factorization, pointwise soundness, and root-set exhaustion.

This organization is intentionally relational. A later executable algebraic- number layer may compute the certificates, but the mathematical formula is not made dependent on a particular computer algebra system's analytic square-root or fractional-power convention.

13 What is proved, and what remains

13.1 Paused Lean checkpoint: 11 August 2026

The project is paused at this commit. The following vocabulary is used in this section.

Committed and kernel-checked
: The named source is in Git, its focused Lean target was rebuilt from the current source, and representative exported theorems were checked with `#print axioms`. Standard logical dependencies such as `propext`, `Classical.choice`, and `Quot.sound` are not treated as proof gaps.

Source or local-build checkpoint
: The mathematical construction is present, and in some cases the root file has elaborated, but the complete current dependency closure and public aggregate have not both been rebuilt. Such an item is not counted as a completed public theorem.

Open
: A proof bridge, bounded certificate frontier, fresh dependency build, or executable construction is still required.

The principal *committed and freshly checked* results at the pause are:

1. The reusable algebraic endgames are proved: inverse Fourier reconstruction gives roots from the four cyclic relations, and the five Vieta identities give the exact linear-factor product and root exhaustion when the polynomial is not identically zero. The established degree-at-most four solvers return explicit radical expressions and have soundness and exhaustiveness theorems. These statements should not be confused with an executable complete quintic solver.

2. `LazardGeneralResolventCriterion.lean` proves the corrected orbit-resolvent criterion. Containment in the displayed stabilizer implies a base-field root without a separability premise. Conversely, an injective specialized orbit (equivalently the required separability condition) gives containment in a *conjugate* stabilizer, which is the label-invariant conclusion.

3. `LazardGeneralResolventExplicit.lean` constructs the rename action, universal coset values and their specialization, derives Galois equivariance from an exact ordered-root presentation, proves coefficient descent, and produces a base-field resolvent whose scalar extension is the orbit product. These are derived objects, not supplied equivariance or coefficient certificates.

4. `LazardGeneralResolventThetaAdapter.lean` proves the exact F_{20} stabilizer of the ten-term theta invariant and identifies its six-coset orbit product, after scalar extension, with the existing Frobenius–Dummit/Lazard sextic. This closes the generic identification; it does not by itself freshly rebuild all specialized integer coefficient computations.

5. The unconditional alternating-resolvent assertion in characteristic three has a committed kernel-checked counterexample. More generally, the formal audit replaces false printed statements by corrected theorems or explicit counterexamples rather than silently strengthening hypotheses.

6. `LazardGeneralResolventConjugacyCounterexample.lean` gives the finite S_5/F_{20} witness to the labelling error. A conjugate of the standard F_{20} fixes the selected non-base coset although it is not contained in the displayed standard subgroup. Its terminal theorem has only the standard assumptions `propext`, `Classical.choice`, and `Quot.sound`.

7. One previously intractable coefficient leaf of the executable Dummit decision procedure is now a bounded, deterministic certificate graph. Table 0 term 125 is decomposed into 358 Lean modules, with 143 ordinary `decide` leaves. All 358 modules were rebuilt; the terminal theorem has assumption report `[propext]` and its state-zero leaf has none. There is no `native_decide`, admission, unsafe implementation, or external proof oracle in this graph. This completes one frontier node, not the full coefficient table.

The following substantial pieces are *source or local-build checkpoints*, not yet a single clean public theorem graph:

1. `LazardGeneralResolventMinpoly.lean` is a buildable prefix checkpoint. It proves the fraction-action/rename stabilizer identity, the stabilizer of a simple fixed-field adjunction, fixed-field generation from an exact rename stabilizer, and the converse generation criterion. Those five exported endpoints have only standard logical assumptions. The product reindexing needed for the ordinary minimal-polynomial identity still has a quotient-`Fintype` instance mismatch; that theorem and its three dependents are preserved verbatim in labeled comments, with no `sorry`, axiom, or placeholder. The result is therefore not yet the promised minimal-polynomial theorem.

2. The Dummit coefficient implementation has been rewritten so that the large finite identities are intended to be checked by bounded ordinary kernel reductions. Hundreds of generated table/root modules still await their serial build, and the top `ComputableDummitCoefficients` artifact is stale relative to that rewrite. Consequently downstream assumption reports can still display four legacy `native_decide` constants even though the current source no longer contains them.

3. The root-origin Lazard chain constructs a labelled complete root tuple, derives the cyclic Fourier and Vieta identities, checks the corrected displayed numerators, and feeds them to inverse Fourier reconstruction. The denominator-safe alternate construction coherently recovers all four nonzero Fourier modes without dividing by E. Focused root files and the explicit $E = 0$ Gaussian-integer example have elaborated locally, but they were checked against the stale top Dummit artifact and therefore require a fresh transitive rebuild before being promoted here.

4. The Section 7 aggregate packages pointwise soundness, exact factorization, and exhaustion from its root-origin data. Its own source elaborates, but its present cached assumption report inherits the four stale Dummit constants just described.

5. The classical Gaussian-rational dispatcher specifies that every vector it returns is an exact exhaustive root vector and that a nonzero input has such a vector exactly when all roots are radical-solvable. The solvable genuine-quintic branch uses classical choice of a proof-carrying solution; no exact algebraic-number program presently computes those expressions. Its changed source and the public import/audit surface have not received the final clean build.

6. The modular audit of printed Theorem 2 has a concrete regular $C_6/\mathbb{F}_3$ contradiction under development. The formal lower bound is 17 independent quotient classes, not a proof that the quotient has exact dimension 17. The corrected nonmodular theorem uses invertibility of the group order as a sufficient Reynolds-averaging hypothesis.

7. The polynomial-realized fixed-subgroup counterexample is written, but its exact-root and scalar-resolvent dependencies run through the unfinished Dummit closure. It must not be advertised merely on the strength of an old cached artifact.

8. Rocq/Coq contains many parallel proofs and bounded-certificate drafts. That work has been checkpointed separately and is explicitly postponed; this paused status and the restart plan below concern Lean first.

Accordingly, this repository does *not* yet contain a complete formal proof of every claim in Lazard's paper. Such a claim would in any event be incorrect: some printed claims (including the literal subgroup conclusion of Theorem 1 in an arbitrary labelling, the arbitrary-characteristic version of Theorem 2, and the minimality assertions in Theorems 3 and 4 under the paper's Definition 1) require correction or are false. Nor is there yet a Lean function which, for every Gaussian-rational quintic, computes either explicit radical expressions or a proof of nonsolvability. What exists is a mixture of an honest primitive-recursive solvability decision proof, a noncomputable proof-carrying expression specification, and a growing kernel-checked transcription of Lazard's explicit construction.

13.2 Ordered next steps

When work resumes, the shortest dependable path is:

1. Continue the bounded Dummit certificate frontier immediately after table 0 term 125, building one generated module or small bounded aggregate at a time. Finish all four coefficient tables and the root-state certificates.

2. Rebuild `ComputableDummitCoefficients` and its public decision theorem from that source closure, then verify that the four stale `native_decide` constants disappear from `#print axioms`.

3. Rebuild, in dependency order, the scalar resolvent criterion, root ordering, root-reduction bridge, Section 7 aggregate, coherent alternate tower, and the actual-root $E = 0$ example. Repeat the endpoint assumption audits only after this fresh chain exists.

4. Resolve the quotient-`Fintype` reindexing in the universal minimal-polynomial module, restore its three dependent endpoints, and then rebuild the realized polynomial counterexample. Keep the universal rational-function minimal polynomial distinct from specialized sextic minimality, which can fail when specialization collides.

5. Complete and kernel-check the modular Theorem 2 contradiction and the corrected non-modular invariant-module theorem, retaining only the proved quotient-rank lower bound.

6. Finish the bounded coefficient certificates for the displayed P_2 and P_3 Fourier numerators, then integrate the root-derived Fourier/Vieta graph and both the standard and alternate reconstruction paths.

7. Integrate and rebuild the Gaussian dispatcher/completeness specification and the root/public audit modules. If executable radical output is required, add a separate exact algebraic-number witness-selection layer rather than treating classical choice as computation.

8. Run the complete public Lean build and audit every exported endpoint for `sorry`, new `axioms`, `native_decide`, `implemented_by`, `unsafe`, and `exact_no_check`. Only then update this checkpoint from "source/local-build" to "completed public theorem."

Proof scripts exist for both formula-independent endgames. For components $P_1, \ldots, P_4$ satisfying `FourierRelations`, `FourierRelations.inverseFourier_root` proves that each inverse Fourier value annihilates the depressed quintic, and `solveGeneral_root_of_fourierRelations` transports that result through the Tschirnhaus translation. For five values satisfying `FiveRootRelations`, `FiveRootRelations.factorization` proves the exact polynomial product, and `FiveRootRelations.exists_eq_of_eval_eq_zero` proves root exhaustiveness when the leading coefficient is nonzero. These proofs use only symbolic field identities. The depressed specialization now factors the repeated endgame into `DepressedFiveRootRelations.eval_root`, `DepressedFiveRootRelations.exists_eq_of_eval_eq_zero`, and `DepressedFiveRootRelations.eval_eq_zero_iff`. The Lean Section 7 aggregate exposes the resulting pointwise soundness and all-root exhaustion as fields derived from its root-origin Vieta data, matching the corresponding Coq aggregate. Coq now factors the same general implication through `lazard_depressed_vieta_root`, `lazard_depressed_vieta_complete`, and `lazard_depressed_vieta_root_iff`; neither package accepts those conclusions as premises.

The source-level general-rational composition is now explicit in both provers. In Lean, the older printed-quotient route uses `completelySolvableByRadicals_polynomial_iff_depress_polynomial`, which transports semantic solvability through the affine root bijection, and `exists_radicalFormula_completeRootVector_of_completelySolvableByRadicals` composes that transport with the rational-resolvent criterion and the five-root Lazard output. The separate corrected theorem `exists_general_commonCompositum_coherentAlternate_` `completeRootVector_of_completelySolvableByRadicals` uses the common-compositum four-fifth alternate construction and has no $E \neq 0$ proof dependency. The shared affine transport and resolvent-root consequence now live in `LazardQuinticGeneralSolvabilityTransport.lean`, while the two shared Figure–3 scalar-transport lemmas live in `LazardQuinticInvariantSystemMap.lean`. A recursive source import audit of the corrected endpoint reaches those foundational modules but not the standard solvability/end-to-end wrappers, printed Section 7 root-origin modules, or the E-nonzero module. In Coq, `LazardQuinticGeneralRationalEndToEnd.v` defines the literal nonmonic polynomial and its monic depressed translate, proves the polynomial identity, root bijection, irreducibility transport, and explicit `algterm rat` translation, then composes the denominator-safe common-compositum adapter. Its resolvent, explicit-radical, and original-Galois-solvability entry points retain only the necessary nonzero leading coefficient and irreducibility hypotheses. The corrected Lean and Coq packages both record injectivity of the five translated values and the exact leading-coefficient-scaled linear-factor product for the original polynomial, so both endpoints are multiplicity-sensitive rather than merely exhaustive at the level of root support. These newly registered endpoints remain source-level until the complete Lean and Coq public builds accept them.

The raw `InvariantRelations` and `RadicalCertificate` interfaces are insufficient on their own because they admit extraneous solutions. Accordingly, the older `FormulaSound`, `RationalFormulaSound`, `FormulaFactors`, and `RationalFormulaFactors` definitions are not asserted: their present hypotheses omit necessary evidence. The newer root-origin modules derive the cyclic Fourier and Vieta identities from a complete ordered root tuple and therefore avoid that defect. The remaining work is integration and kernel validation of that derivation, not another request that the caller postulate the identities.

The proof-carrying Gaussian quintic dispatcher requires both relation families in every Lazard witness, packages that witness as an exact `CompleteRadicalSolution`, and proves completeness for every vector it returns. More generally, for a Gaussian-rational tuple c with `c.Nonzero`, `solve_hasRootVector_iff_completelySolvableByRadicals` proves

$$\texttt{(solvec).HasRootVector} \iff \forall x \in \mathbb{C},\ c(x) = 0 \Longrightarrow x \in \texttt{solvableByRadQC}.$$

The all-zero tuple is excluded because its root set is infinite. Whenever `rootVector?` returns a vector, the stronger theorem `solve_eval_eq_zero_iff_returnedVector_contains` characterizes its entries exactly as the complex roots; the stored product identity also accounts for multiplicity.

This dispatcher theorem is an existence specification, not an executable decision procedure. For a semantically solvable genuine quintic, Lean uses the complex root multiset and classical choice to build a `CompleteRadicalSolution`; it also classically tests the propositions that the Lazard and fallback packages are inhabited. Thus automatic construction of a Lazard witness and executable selection of radical expressions remain absent even though the finite-output equivalence is proved.

The Section 5 degree audit also needs two distinctions which are easy to lose in an informal argument. First, the claims that the least degrees of exact C_5- and D_5-stabilizer invariants are three and two concern arbitrary polynomial degree, not only a preselected homogeneous component; the paired sources now obtain the arbitrary statements by homogeneous decomposition. Second, a collision modulo C_5 need not remain a collision modulo D_5. Separate Lean and Coq theorems therefore prove repeated factors in the literal S_5/C_5 and S_5/D_5 orbit products. Finally, since Lazard defines "always separable" by quantifying over irreducible inputs, the new paired witness modules prove $X^5 - 2$ irreducible by Eisenstein, construct its distinct five-root tuple in an algebraically closed extension, prove that its exact monic factorization is the product of the five displayed linear factors, prove the corresponding root-membership iff/exhaustion statement, and instantiate both nonseparability results there. These statements remain source-level until the focused and public kernel builds accept them.

This necessary degree bound is logically independent of Lazard's next sharpness claim. The paper exhibits the degree-five products T' and U' and epsilon, and also observes that $T'U'$ is a square root of the discriminant. The paired root-radical sources now define the ordered Vandermonde product

$$\Delta = \prod_{0 \leq i < j < 5} (x_i - x_j)$$

and prove the orientation-sensitive identity $T'U' = -\Delta$, hence $(T'U')^2 = \Delta^2$, as well as nonvanishing for distinct roots. The paired relative-resolvent sources package this intermediate invariant on $D_5 < F_{20}$, identify the signed quotient value with the actual root action for every representative, and derive separability; irreducible-rational wrappers derive distinctness and $2 \neq 0$ internally. Here "discriminant" is the paper's root discriminant Δ^2; no coefficient-level library discriminant identity is being silently substituted.

The paper further says that the two-valued relative resolvents for $C_5 < D_5$ and $D_5 < F_{20}$ are always separable. The paired developments already prove the relevant root products nonzero and record their sign changes under the nontrivial quotient representatives. The new paired `LazardSection5DegreeFiveRelativeResolvents` modules now package those ingredients as literal two-coset orbit products. They prove the two cosets and their inequality internally and derive $t \neq -t$ from $t \neq 0$ and the explicit hypothesis $2 \neq 0$; no separability or coset-distinctness certificate is supplied by the caller. New rational-quintic wrappers for T' and U' also derive duplicate-freeness of the ordered roots and $2 \neq 0$ directly from irreducibility and characteristic zero, so their paper-facing statements do not leave a root-separability certificate with the caller. Each development also exposes one combined endpoint for all three candidates on a single irreducible depressed rational quintic/selected ordering. Coq additionally has an arbitrary-rational monic depressed wrapper: the existing denominator adapter supplies a complete rescaled root tuple, the new source proves that epsilon is homogeneous of degree five under root dilation, and nonzero denominator cancellation transports both injectivity and canonical epsilon nonvanishing back to the caller's polynomial. Thus its combined endpoint no longer stops at the integral-coded coefficient representation and accepts no epsilon or separability certificate. These wrappers remain source-level until the focused and public kernel builds accept them. This composition is neither used by nor silently assumed in the proof that every degree-below-five candidate fails. The combined epsilon endpoints still take a primitive fifth root in the chosen ambient field; they do not infer that such a root lies in the bare quintic splitting field. A premise-free existence wrapper must instead use a common cyclotomic or algebraically closed extension. Coq now supplies that composition over `algC`: it maps the complete rational root tuple through the canonical embedding, constructs a primitive fifth root there, transports epsilon-product nonvanishing, and concludes all three relative resolvents are separable. Lean now supplies the matching composition over $\mathbb{C}$: it embeds the canonical splitting-field root tuple, transports injectivity and the elementary-symmetric coefficient identity, uses the explicit two-square-root primitive fifth root, derives epsilon nonvanishing from irreducibility, and concludes that all three relative resolvents are separable. Neither common-field endpoint asserts cyclotomic containment in the bare quintic splitting field.

For reference, the earlier fine-grained source audit recorded the following backlog. It is more expansive than the Lean-first restart order above. The existence of source files for these items must not be confused with acceptance by a current public kernel build:

1. kernel-check the field-generic Lean Fourier soundness, root-origin inversion, derived five Vieta relations, exact polynomial factorization, soundness, and exhaustive-root endpoints under the exact hypothesis $5 \neq 0$, together with the matching Coq Fourier/Vieta graph;

2. finish and kernel-check the symmetric rational-function fixed-field, finite-extension, Galois-group, and minimal-polynomial bridges in both provers;

3. kernel-check the arbitrary-base-field Chapman no-collision endpoints under $5 \neq 0$ and the new splitting-field specializations that derive their root-sum and five-cycle-action hypotheses from a general irreducible degree-five polynomial;

4. kernel-check the corrected exact nonmodular form of Theorem 2 (together with its characteristic-zero convenience wrapper) and the explicit, semantically closed modular $C_6/\mathbb{F}_3$ counterexample to its printed arbitrary-characteristic form, including the proved group cardinality and public no-assumption endpoint;

5. kernel-check the Lean generic quadratic and Cardano formulas, the exceptional $p = 0$ branch, both literal cubic root resolvents and their simultaneous action, the corrected S_3 subgroup statement and its order-two counterexample, and the characteristic-three obstruction, then rebuild both enlarged public audits. The corresponding focused Coq Section 4 modules are kernel-green with closed assumption reports. Also kernel-check the exact Sect. 3 criterion $\mathrm{Sep}(X^2 - \Delta) \leftrightarrow (2 \neq 0 \wedge \Delta \neq 0)$ and the closed irreducible-inseparable $X^3 - t$ counterexample over $\mathbb{F}_3(t)$;

6. kernel-check the complete five-class transitive-subgroup classification, the conjugacy correction to Theorem 1, including both its finite S_5/F_{20} witness and the explicit irreducible-rational polynomial realization with exact roots, genuine splitting-field Galois action, separable base-root resolvent, corrected conjugate containment, and failed displayed containment; kernel-check the exact C_5/D_5 stabilizers and the arbitrary-polynomial minimum-degree theorems, the separate literal S_5/C_5 and S_5/D_5 low-degree collision theorems, and their irreducible $X^5 - 2$ instantiations, and the newly packaged displayed degree-five T', U', $T'U'$, and epsilon literal two-coset relative resolvents whose separability is derived from the existing nonvanishing and sign identities under the exact hypothesis $2 \neq 0$, including the combined irreducible-rational endpoints which derive root injectivity internally;

7. kernel-check all four literal Figure 3 identities, their four-by-four coefficient matrix, and the irreducible-rational determinant-nonvanishing chains in both provers. The new Coq rational transport proves the exact diagonal matrix scaling and determinant weight D^{30}. Canonical Vieta and the denominator adapter identify the rescaled complete-root coefficient record with the scalar extension of the original literal rational record (p_3, p_2, p_1, p_0), while the matrix and determinant commute with that extension. The resulting base-rational and root-level terminal theorems derive determinant nonvanishing and invariant uniqueness for an arbitrary rational monic depressed irreducible quintic without a caller-supplied separability or determinant certificate. Thus the former source-scope mismatch is closed, but the gate remains open until this transport and its canonical determinant dependencies pass the focused and public kernels;

8. kernel-check the source-level literal displayed Gröbner basis for arbitrary degree and every admissible block order, and the corresponding intrinsic-to-combined leading-term descent. Both provers now derive the support, division/normal-form, and leading-term data internally rather than accepting them as certificates. Lean's literal Lemma 2 bridge currently assumes characteristic zero, while Coq exposes the precise nonzero image of $|G|$. Also kernel-check both direct Molien numerator-series bridges and the concrete six-element bases through the public imports. The Coq source now proves coefficientwise that multiplication by the symmetric denominator gives $1 + t^4 + t^5 + t^6 + t^7 + t^8$ and identifies this polynomial with the actual six displayed basis degrees; this closes the former source gap, but the new dependency graph still awaits its focused and public kernel builds;

9. kernel-check the corrected P_{22}, all remaining displayed numerator identities, and the root-derived Fourier/Vieta aggregate in both provers;

10. kernel-check the newly registered general-rational Lean/Coq wrappers that connect original-polynomial solvability, depression, rational-resolvent root selection, F_{20} Galois action, coefficient descent, and the denominator-safe four-fifth radical tower in a common overfield containing the roots and a primitive fifth root;

11. kernel-check the explicit counterexamples to every false printed claim, including Theorems 3 and 4, and validate the corrected subgroup/fixed-field/Kummer degree statement that replaces their invalid minimality conclusions;

12. after Lean is complete and Rocq work resumes, run both public builds, print assumptions for the exported theorems, and audit for admissions, new axioms, and unsafe computation escapes.

Separately, classical certificate selection can be replaced by an executable exact algebraic-number implementation if computational output, rather than a formal existence theorem, is desired. That engineering task is not a substitute for any of the mathematical kernel gates above.

The generic Vieta theorem already gives factorization and exhaustiveness from its certificate; it does not assert pairwise distinctness, which would require separability. The freshly audited committed modules listed in the dated checkpoint contain no `sorry`, new `axiom`, `native_decide`, unsafe implementation, or opaque external proof oracle. That statement is deliberately not extended to a stale cached dependency closure before the final rebuild and assumption audit.

14 Regression evidence and source cautions

The wider Smithereens review found useful future regression cases, but also showed why notebook output is not a proof oracle.

- The corrected zero-T branch is exercised by $X^5 - 10X^2 - 10X - 6$, for which the audited source gives $i_4 = 30$.
- An older direct notebook records a failure on $X^5 - 5X - 12$ when it omits an adequate branch fallback. This is a valuable all-five-roots case.
- Reshetnikov's strongest stored exact harness checks five values, substitution residuals, and distinctness for six polynomials, including the very large target polynomial from the Stack Exchange question.
- Some other notebook runs abort after ten cases, some use `PossibleZeroQ`, and the independent package checks high-precision residuals without proving symbolic identities or complete root coverage.
- The label of $X^5+20X+32$ as F_{20} in one newer document is incorrect; its group is dihedral. It should not be used as an F_{20} classification fixture.
- Dummit's printed examples have later corrections, and his original factorization proof needs Chapman's specialized-distinctness lemma.

Consequently the committed Lean module uses source formulas and algebraic certificates, not notebook success labels. Future regression computations are useful checks against transcription errors, but kernel-checked symbolic identities will remain the correctness criterion.

15 Reproducibility and audit

From the repository root, the freshly checked focused targets at this pause include

```
lake env lean \
  Algebra/PolynomialFormulas/Lean/PolynomialFormulas/LazardQuintic.lean
lake env lean \
  Algebra/PolynomialFormulas/Lean/PolynomialFormulas/LazardQuinticFourier.lean
lake build +PolynomialFormulas.LazardGeneralResolventCriterion:olean
lake build +PolynomialFormulas.LazardGeneralResolventExplicit:olean
lake build +PolynomialFormulas.LazardGeneralResolventThetaAdapter:olean
lake build +PolynomialFormulas.LazardGeneralResolventConjugacyCounterexample:olean
lake build +PolynomialFormulas.LazardGeneralResolventMinpoly:olean
lake build \
  +PolynomialFormulas.ComputableDummitKernelCertificateTable0Term125Certificate:olean
```

The Gaussian completeness targets and the full `PolynomialFormulas` aggregate belong to the ordered restart plan; listing their eventual commands would not be evidence that the present worktree has passed those builds. The document is rendered by first changing to its directory and then running the following command three times (for the table of contents, citations, and cross-references):

```
cd Algebra/PolynomialFormulas
pdflatex -interaction=nonstopmode -halt-on-error \
  LazardQuinticFormalization.tex
```

For the present contribution, an audit should additionally verify:

1. no occurrence of `sorry`, `admit`, or a new `axiom`;
2. the displayed corrected terms occur as `8*c.p^3*c.q` and `70*c.q^3`;
3. `correctEpsilon` precedes construction of T and U;
4. `solveDepressed` uses one stored P_1 and one primitive ω;
5. `LazardWitness` stores the Fourier relation family, while the generic development derives both the Vieta relations and the polynomial identity;
6. the public finite-vector equivalence retains its `c.Nonzero` hypothesis and is documented as classical and noncomputable; and
7. the PDF is built from the committed LaTeX source without errors.

References

[1] Daniel Lazard, *Solving Quintics by Radicals*, in Olav Arnfinn Laudal and Ragni Piene (eds.), *The Legacy of Niels Henrik Abel*, Springer, 2004, pp. 207–225. https://doi.org/10.1007/978-3-642-18908-1_6.

[2] David S. Dummit, *Solving Solvable Quintics*, Mathematics of Computation 57 (1991), no. 195, 387–401.

[3] Robin Chapman, *Note on Solving Solvable Quintics*, one-page correction note hosted with Dummit's source materials. https://site.uvm.edu/ddummit/files/2021/04/Solving_Solvable_Quintics_note_by_Robin_Chapman.pdf.

[4] Ben Blum-Smith and Sophie Marques, *When are permutation invariants Cohen–Macaulay over all fields?*, Algebra & Number Theory 12 (2018), no. 7, 1787–1820. https://doi.org/10.2140/ant.2018.12.1787.

[5] Vladimir Reshetnikov and Valeriy Zapodovnikov, *Solving quintic in radicals*, Mathematica Stack Exchange question 133550 and answers, 2016–2025. https://mathematica.stackexchange.com/questions/133550/solving-quintic-in-radicals.